Stair Fusion Network With Context-Refined Attention for Remote Sensing Image Semantic Segmentation

Jia Liu, *Member, IEEE*, Wenyi Hua, Wenhua Zhang, *Member, IEEE*, Fang Liu, *Member, IEEE*, and Liang Xiao, *Senior Member, IEEE*

***Abstract*— Semantic segmentation of remote sensing images is essential in various fields, such as Earth resource census, environmental pollution monitoring, and land use planning. The segmentation performance has been significantly improved recently with the development of deep learning. However, there are still some challenges in dealing with remote sensing images. One of the main issues is that features within the same category in remote sensing images could vary significantly, while features between different categories could be more similar, leading to confusion in segmentation. Moreover, the presence of large shadow areas narrows the feature differences between categories, making segmentation even more difficult. To address these challenges, one way is to leverage contextual and multiscale information for accurate segmentation. As a consequence, in this article, we propose a stair fusion network with context-refined attention (SFCRNet). A context-based attention embedding module is proposed to enhance the representation of the processed features by utilizing the context to maximize information retention in the channel and spatial dimensions. It can retain the information on the original channel and the association between it and other channels. Furthermore, we present an SFN, where a stair-shaped architecture and corresponding fusion module are designed to ensure that rich semantic information from high-level features is continuously transmitted to low-level layers. The experimental results on three datasets demonstrate the effectiveness of our proposed method.**

***Index Terms*— Attention mechanism, remote sensing images, semantic segmentation, stair fusion.**

I. Introduction

With the development of remote sensing imaging technology, a large volume of observation data with high resolutions can be obtained which provides requirements and challenges for various remote sensing image processing methods. Semantic segmentation is one of the most fundamental tasks in remote sensing image processing. Generally, semantic segmentation is defined as the classification of each pixel. Extended to the whole image, it is the category division of a specific area that forms an intuitive sense of the target [1]. Thus, semantic segmentation of remote sensing images is widely employed in many practical tasks, such as Earth resource census [2], environmental pollution monitoring [3], [4], and land use planning [5], [6].

Semantic segmentation in natural images has made considerable progress, with deep learning-based methods performing far better than traditional methods. The proposal of fully convolutional networks (FCNs) [7] laid the foundation for the semantic segmentation task in deep learning, and most of the subsequent work inherited the modeling of FCN for the semantic segmentation task. In the subsequent work, the encoder–decoder structure exhibited powerful semantic segmentation capabilities and was adopted by most semantic segmentation networks [8], [9], [10]. The encoder extracts the features of the input images and inputs them into the decoder. The decoder restores semantic information of the features to obtain the refined segmentation result. However, due to the nature of the convolution operation itself, it is hard for convolutional neural networks to establish long-distance dependencies. To solve this problem, Zhao et al. [11] proposed a pyramid pooling module (PPM), which accomplishes establishing and fusing long-distance dependencies by pooling features of different sizes and stacking them together. Chen et al. [12] proposed atrous spatial pyramid pooling (ASPP), which gradually expands the receptive field of convolution by using different dilation ratios to better learn contextual information.

For remote sensing images, semantic segmentation is more challenging than in natural images. One of the main points is that objects within the same category may differ significantly in features, while objects between different categories may have more similar features [13], [14], [15]. Humans can make a judgment on the precise classification of objects by extensively observing the surrounding information, i.e., contextual information. However, due to the locality of convolution, it judges the category mainly by learning the object's texture, and the lack of contextual information will lead to wrong classification. Second, there are many shadows in remote sensing images [16], [17]. Large shadows will lead to little feature distinctions among different categories and increase judgment difficulty. Small shadows will truncate the continuous features within a single object and lead to discontinuity of segmentation results. Among the features extracted by the

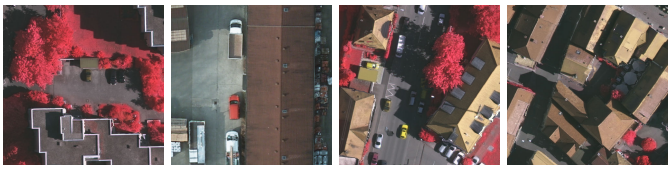
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The authors are with the School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing 210094, China (e-mail: whzhang@njust.edu.cn).

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(a) (b) (c) (d)

Fig. 1. Examples from ISPRS Vaihingen dataset and ISPRS Potsdam dataset illustrate the challenges in semantic segmentation of remote sensing images. (a) Top of building is similar to impervious surface. (b) Difference between the three cars in the figure is significant. (c) Cars in the shadows are more challenging to spot than cars in the sun. (d) Shadow divides the impervious surface into several parts, but they are actually connected.

convolutional neural network, the high-level features have rich semantic information and better continuity. Nevertheless, there is also the possibility of false continuity, and the smaller resolution of features will lead to the uncertainty of the boundary. The low-level features have rich spatial information and more accurate object boundaries, but the continuity needs to be improved and easily affected by the shadows mentioned earlier. As a consequence, it is critical to effectively fuse the low and high-level features. Fig. 1 shows some specific examples of the above-mentioned problems. Through the above-mentioned analysis, designing efficient architectures and objective functions to fully extract and use contextual [14], [18] and multiscale information [19], [20] is one of the ways to solve the difficulties.

To use the contextual information, the most prominent approach is the attention mechanism. The use of attention mechanisms in computer vision could be broadly divided into two categories. One is the attention mechanism based on information aggregation, where the features are first aggregated according to the dimension of attention required. Then, some linear transformations and nonlinear activations are performed on the aggregated information to obtain the attention score. The other is the attention mechanism based on the similarity between units, which constructs a similarity association between each pair of units by matrix multiplication to obtain the attention score. For the former, for example, Hu et al. [21] proposed SEBlock, using average pooling and a multilayer perceptron with a squeeze-and-excitation (SE) structure to obtain the attention score in the channel dimension. The convolutional block attention module (CBAM) [22] further improves the SEBlock by introducing max pooling to enrich the features, and an attention mechanism in the spatial dimension is proposed by changing the dimension of information aggregation. ST-UNet [23] goes one step further by introducing soft pooling while aggregating the information into two dimensions, height and width, respectively. For the latter, for instance, the nonlocal block [24] constructs the relationship between any two pixels at any moment. Subsequently, DANet [25] eliminates the temporal dimension and constructs the relationship between pixels and the relationship between channels on images. MANet [26] and reduced global attention module (RGAM) [27] reduce both space and time complexity by mathematically simplifying and introducing cheap linear operations to reduce the size, respectively.

To fully employ the multiscale information, feature fusion is usually used. Fusing features at different scales obtained in the encoding stage can enhance the features' expression capability. U-Net [28] and SegNet [8] continuously upsample features and fuse them with features at larger scales through skip connections and reversing the encoding stages so that the final features have sufficient semantic and spatial information simultaneously. To facilitate the operation and simplify the computation, feature pyramid networks (FPNs) [29] unify the number of channels by convolutional operation and then use elementwise summation for information fusion. Based on FPN, UpperNet [30] concatenates the fused features for further information fusion. Unlike the general layer-by-layer feature fusion, DC-Swin [31] uses the fusion method of skipping layers to preserve semantic information as much as possible.

Existing attention mechanism and feature fusion methods can well capture the contextual and multiscale information, respectively. But there are still some defects. In the attention based on information aggregation, only simple operations, such as average pooling and max pooling, are usually used [21], [22], [32]. The aggregated results do not represent the original information well, and the differences among the channels in features are weakened as a result. Moreover, due to the problem of a small convolutional receptive field, it is difficult to further improve the attention effect in the spatial dimension. In addition, although the attention based on similarity modeling has powerful representation capability, the spatial and computational complexity is too high to be efficiently used for high-resolution features. Therefore, in this article, we design a context-refined attention module (CRAM) to refine the information aggregation process. First, a set of weights is generated from the input features in the direction of information aggregation. Then, information aggregation by multiplying and summing is used to replace the simple mean or maximum value in existing methods. The aggregated information will undergo a series of linear transformations and nonlinear activations to obtain the final attention score. Meanwhile, to alleviate the typical small convolution kernel's deficiency in the receptive field, we adopt the RepLKConv proposed by Ding et al. [33] to help improve the performance of large convolution kernels by residual branching with reparameterization instead of simply increasing kernel size. In feature fusion, existing methods usually have only one path from the fusion of high-level features to low-level features, and the rich semantic information in the high-level features will be continuously weakened in the fusion and cannot be well transferred to the output features [8], [19], [28]. Based on the goal of allowing the information in high-level features to be better passed downward, we propose a stair fusion network (SFN), which allows the information in high-level features to be passed downward several times by setting multiple paths. Also, we design a feature fusion module in the network to strengthen the feature fusion effect at adjacent scales as much as possible so that the final output features can have sufficient semantic and spatial information. With the CRAM, SFN, and the proposed feature fusion module, the SFN with context-refined attention (SFCRNet) is proposed in this article. We have conducted experiments on three datasets, and the results show that our method outperforms recent

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state-of-the-art methods. The main contributions of this article can be concluded in the following points.

1) We propose CRAM for selective attention to feature channels and spatial dimensions. The information aggregation process in the attention mechanism is redesigned, and the receptive field is enhanced by using large convolution kernels. CRAM can process channel and spatial information at the same time. Different channel information has an effect on shadows, and spatial information can maintain the structural information of the target.
2) We design a network SFN that can fully integrate multiscale features. Semantic and spatial information can be better integrated through multiple feature fusion and information transfer from deep to shallow layers so that the segmentation result can accurately judge the category and have a clear and definite edge and spatial structure.
3) By carefully considering the connections and differences between features of different scales, we propose interaction and complementary feature fusion block (ICFFB), which enhances the ability of SFN feature fusion.

The rest of the article is organized as follows. Section II will elaborate on related work. Our proposed method will be introduced in detail in Section III. In Section IV, we will conduct related experiments on SFCRNet, including a comparison with recent state-of-the-art methods. The conclusion will be given in Section V.

II. Related Work

A. Motivation

Semantic segmentation of remote sensing images faces new challenges compared with natural images. There are high similarities between different categories, and there will also be relatively significant differences within the same category. The influence of superimposed shadows reduces the variance inside the shadows and increases the difficulty of semantic segmentation of remote sensing images. Assisting judgment through contextual information and integrating semantic and spatial information with multiscale features is a feasible way to solve these problems. The CRAM we propose simultaneously processes channel and spatial information through contextual auxiliary information aggregation. Incorporating information from different channels can increase the robustness of shadows, and spatial information can maintain the structural information of the target. SFN uses multiple top-down fusion paths to make the fusion of features more complete and cooperates with ICFFB to consider the differences between features of different scales so that the semantic information can be retained while having rich spatial details to improve the accuracy of the final category determination.

B. Semantic Segmentation

Semantic segmentation is one of the essential tasks of image processing. After FCN [7] proposed the end-to-end training semantic segmentation network method, various methods have sprung up. Among the end-to-end architectures, the encoder–decoder structure has shown prominent performance in semantic segmentation. Training through the pretrained backbone encoder has significantly improved the performance of the segmentation network and has been widely used in subsequent works. The Deeplab series [12], [34], [35] use ASPP to obtain multiscale context information through convolution with different dilation rates. In PSPNet [11], PPM is proposed to obtain context information by pooling features to different scales. UpperNet [30] uses FPN as well as PPM to fuse features to obtain multiscale information to the greatest extent. U-Net [28] and SegNet [8] imitate the reverse process of the encoder in the decoding stage and use jump connections to expand the resolution and improve the spatial details gradually. HRNet [36] uses dense connections to interact between multiscale features in the encoding stage so that the features can maintain high resolution throughout the process. In order to achieve real-time segmentation, BiSeNet [37] uses two different encoding branches responsible for encoding semantic and spatial information, respectively. MaskFormer [38] turns the idea back to the period of traditional methods, which uses the global modeling capability of the transformer to generate a binary classification mask for each category, and finally, integrates the binary classification masks of multiple classes into the semantic segmentation result.

C. Semantic Segmentation of Remote Sensing Images

Remote sensing image processing has new challenges compared with natural image processing, and so many methods have been proposed based on convolutional neural networks as well as the recent transformer focusing on those challenges.

Ding et al. proposed a two-step multiscale training method in [39] to process very-high-resolution remote sensing images. Global features are obtained via downsampling and the original input image is used for detail reduction. PEGNet [40] uses edge information to guide the segmentation process by learning edge features and embedding edge information into each stage of the multiscale decoding network to make the segmentation results more accurate on the edges. DSMFNet [41] uses an additional input, i.e., digital surface model (DSM), through a simple and lightweight fusion module that fuses the information from DSM branches. In UNetFormer, Wang et al. [18] fused the transformer with convolution in the decoding stage, enjoying the powerful global capability of the transformer while fully preserving spatially precise details. He et al. [23] enhanced the global representation of features in the proposed ST-UNet by introducing a transformer-based auxiliary encoder. SSAtNet [42] introduces an attention mechanism into multiscale modules to adaptively refine features and uses a pooling index correction module for multiscale feature fusion to help restore fine-grained information.

In the semantic segmentation of remote sensing images, many works use a U-shaped encoder–decoder structure and skip connections to fuse the multiscale information obtained by the encoder in the decoding stage. However, only one top-to-bottom fusion is insufficient, and most of them use a relatively simple fusion method, which lacks consideration of the representation differences between features at different scales.

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Based on these deficiencies, we propose SFN and ICFFB where a multistage fusion architecture and a fusion method that considers the differences between features are used to obtain better fused features. Moreover, considering that in the existing methods, the pooling operation is usually employed for information aggregation in the attention mechanism, the relation between the original data is easily weakened. Therefore, we propose CRAM, which uses contextual information to refine the information aggregation process and strengthens the attention mechanism's capacity.

D. Attention Mechanism

In recent years, the attention mechanism has been widely used in computer vision, natural language processing, and other fields. In computer vision, unlike typical convolutions that only consider the content within a local receptive field, the attention mechanism focuses more on the acquisition and association of global information. Through global selective attention, long-distance dependencies are established. Hu et al. [21] proposed an SE block in SE networks (SENet), aggregating channel information through global average pooling. Subsequently, through the SE multilayer perceptron, squeeze and excitation are used to obtain the commonality between channels and represent the individuality of different channels. ECANet [32] replaces the global connection by paying attention to the connection between adjacent channels and replacing the multilayer perceptron in the SE block with a convolution which improves the attention mechanism's efficiency and achieves better results. In CBAM [22], spatial attention is proposed by changing the pooling direction to the channel direction. Nonlocal neural networks [24] introduce self-attention into computer vision tasks for the first time, modeling the similarity between any two units in the features. Then, CCNet [43] obtains the context information of the position through two consecutive criss-cross attention modules. DANet [25] establishes dependencies on pixels and between channels by changing the dimension of matrix multiplication in self-attention.

In the works of remote sensing image processing, many attention mechanisms are used. The relation module [14] establishes the dependencies between pixels through matrix multiplication, stacks the original features on top, and embeds the dependencies through convolution. In HMANet [44], the dimension of attention is no longer limited to spatial and channels but is also paid to categories. AFNet [45] designs attention modules for feature fusion in multipath encoders. In LANet proposed by Ding et al. [19], the dependency between space and channel dimensions in attention is considered, features are spatially divided into multiple patches, and channel attention is calculated for each patch separately. SAPNet [46] goes further and proposes a synergistical attention model (SAM), which simultaneously considers the relationship between spatial and channels in self-attention.

Our proposed attention mechanism improves the process of information aggregation in information aggregation-based attention mechanisms. Existing methods usually use the mean or maximum value to aggregate information of a particular dimension in the feature. They cannot well reflect the connection and difference between the information of current dimension in the features. Our proposed method generates a set of weights multiplied and summed with the original data. By strengthening or weakening different positions in the information before aggregation, the representativeness of the information after aggregation is enhanced, and the connections and differences in the original information are reflected as much as possible.

III. Proposed Method

A. Overview

The overall architecture of the SFCRNet is shown in Fig. 2, using the encoder–decoder structure. The encoder uses a lightweight backbone ResNet18 [47] for feature extraction. The decoder consists of our proposed CRAM and SFN, where the fusion block in SFN uses ICFFB for better feature fusion. CRAM aims to refine the attention score by acquiring contextual information and selectively focusing on features. The primary purpose of SFN is to use multiple downward fusion paths to integrate semantic and spatial information fully.

Specifically, a given remote sensing image $X \in \mathbb{R}^{3\times H\times W}$ as input will first be fed into ResNet18 for encoding. After the encoding stage, we will get four-scale features at the four stages of the backbone, with resolutions of 1/4, 1/8, 1/16, and 1/32 of the input image and the corresponding channel numbers of 64, 128, 256, and 512, respectively. Immediately afterward, the four features will undergo a 1×1 convolution with a unified channel number for subsequent processing. Next, the features will first enter the CRAM through selective focus to enhance the representation of the features, and the output size is consistent with the input size. The features after attention enhancement will be sent to the SFN for feature fusion, and then, we will get the feature $X \in \mathbb{R}^{C\times(H/4)\times(W/4)}$, where C is the number of channels after unifying the channels. Finally, we use 1×1 convolution to change the number of channels into the number of categories and use bilinear interpolation to upsample to the same size as the input image to get the final prediction segmentation mask.

B. Context-Refined Attention Module

In remote sensing image semantic segmentation, contextual information will help to establish long-range dependencies, improve the representation of features, and assist semantic judgment. The attention mechanism is one of the main ways to obtain context information, and attention based on information aggregation is widely used in neural networks. Existing considerable attention-related works mainly focus on directly utilizing aggregated contextual information while ignoring the process of contextual information aggregation. However, this process is also vital for the attention mechanism. Excellent information aggregation could more completely preserve the relationship between the elements in the aggregation direction and provide high-quality raw data for subsequent processing. Therefore, we propose CRAM for refining the information aggregation process.

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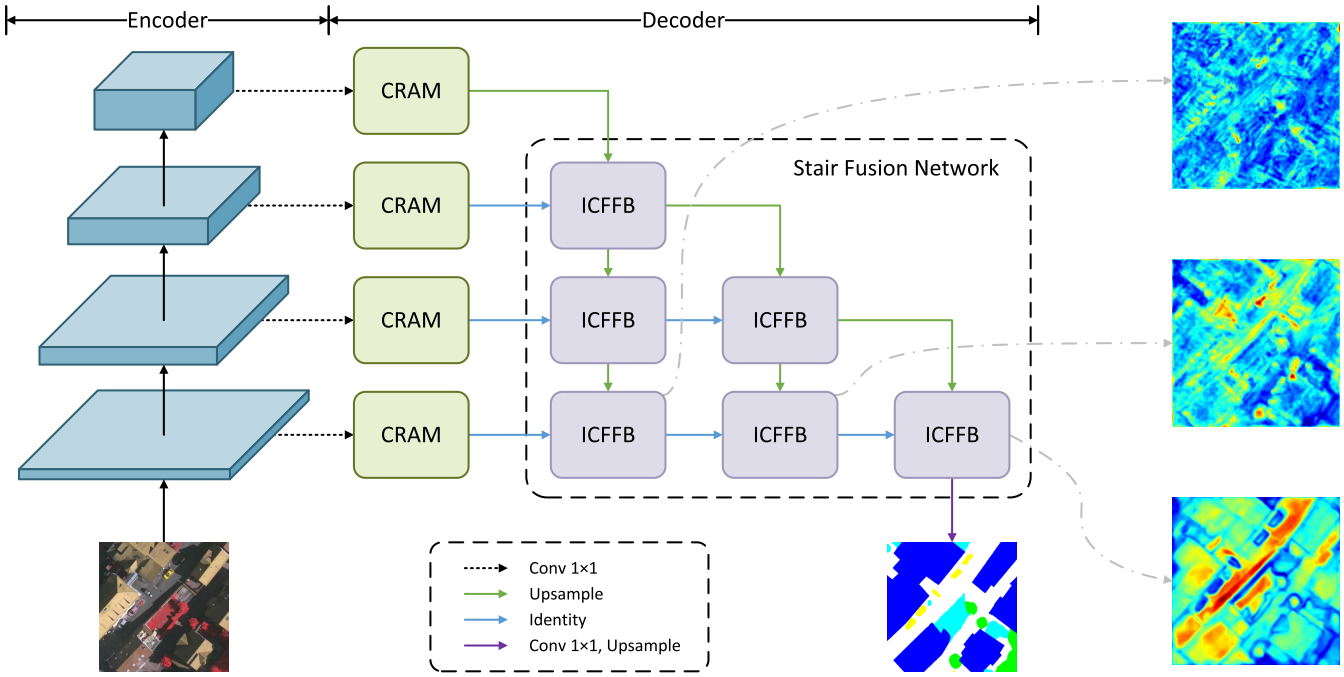


Fig. 2. Overall architecture of the proposed SFCRNet. The three images on the right are the visualization results of the feature maps after the three-stage feature fusion of SFN. With the deepening of the fusion, the semantic information in the features is continuously enriched.

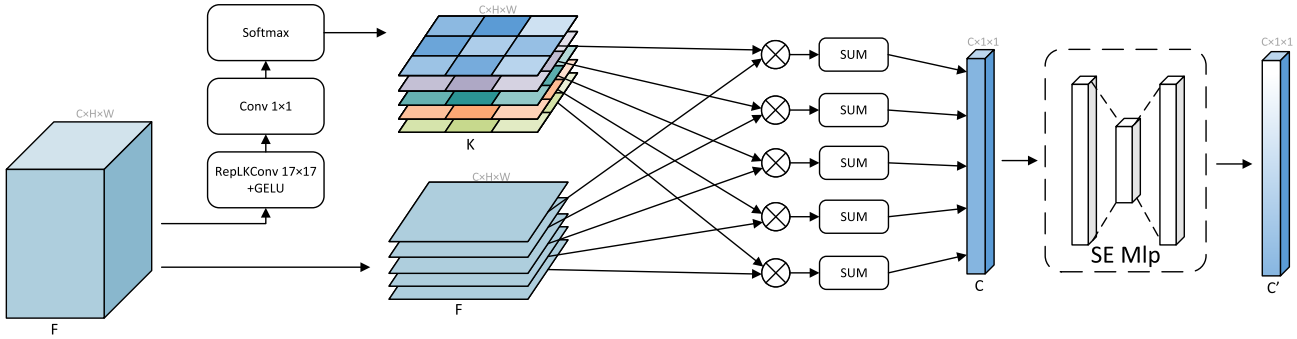


Fig. 3. Structure of CRCA. CRCA utilizes contextual information to assist in refining the aggregation of channel information by generating a set of weights from the input.

CRAM consists of contextually refined channel attention (CRCA) and contextually refined spatial attention (CRSA). CRCA is used to extract channel attention, CRSA is used to extract spatial attention, and the extracted attention scores will be embedded into the input features. Next, we will detail how CRCA and CRSA refine information aggregation through contextual information.

1) Context-Refined Channel Attention: The procedure of CRCA acquiring channel attention is shown in Fig. 3. For the input feature $F \in \mathbb{R}^{C\times H\times W}$, C, H, and W represent the feature's channel, height, and width, respectively. Due to the necessity to quickly establish relationships in ample space, we need a larger convolution kernel, but simply increasing the size of the convolution kernel will cause a bottleneck, and it is even worse than a small convolution kernel. Therefore, we use RepLKConv [33] to utilize large convolution kernels effectively. The specific structure of RepLKConv will be described later. Feature F first passes through the RepLKConv and GELU activation functions to deal with the spatial relationship in each channel. However, since RepLKConv uses depth-wise (DW) convolution, this convolution lacks information exchange between channels. So, then we use 1×1 convolution for processing to deal with information exchange between channels. Then, to avoid excessive numerical values in the following operations, we normalize each channel using softmax to obtain the channel contextual refinement weight $K \in \mathbb{R}^{C\times H\times W}$. Finally, we use the weight K to aggregate the channel information and obtain the aggregated channel information C by multiplying and summing. The information aggregation of the kth channel can be formulated as follows:

$$C_k = \sum_{i=1}^{H}\sum_{j=1}^{W} F_k(i,j) \times K_k(i,j) \tag{1}$$

where i, j, and k are the indexes in height, width, and channel dimensions, respectively. For the obtained channel information C, we use the SE structure of the multilayer perceptron to obtain the final attention score C'. The impact of shadows can be reduced via the channel contextual refinement weight. As shown in the visualized features in Section IV-F4, the features contain features rich in the shaded information as well as features that lack shaded information. The contextual information is integrated into the generation of the weights

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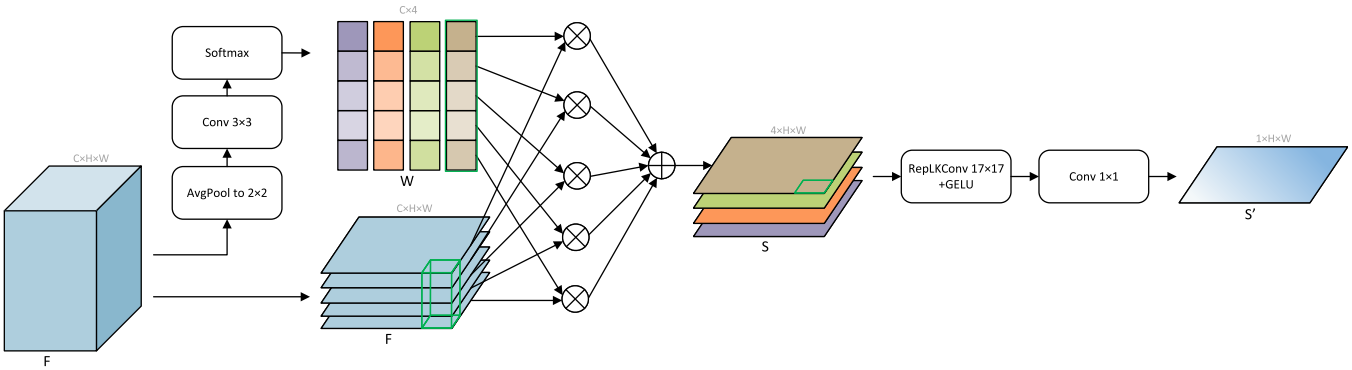


Fig. 4. Structure of CRSA. CRSA utilizes contextual information to assist in refining the aggregation of spatial information by generating a set of weights from the input. Each layer in S is obtained by performing calculations in the illustration with F and each column in W.

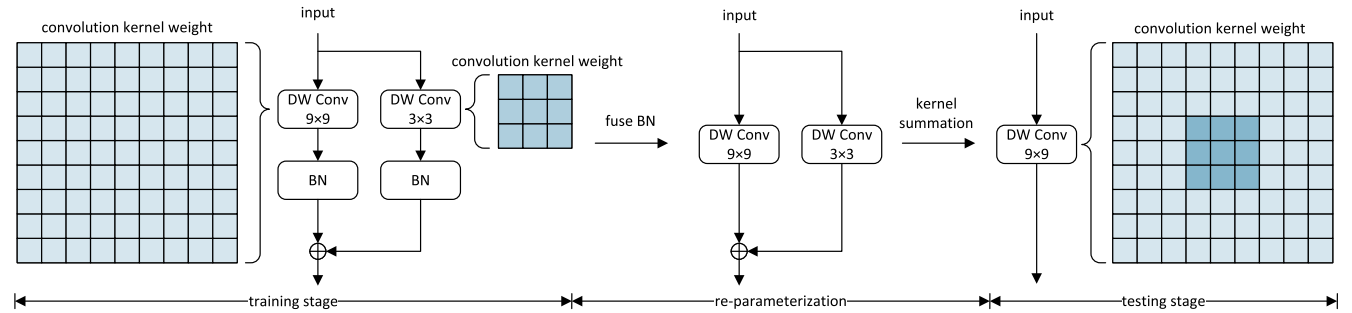


Fig. 5. Structure of RepLKConv, taking 9 × 9 as an example. The left-hand part is the structure of RepLKConv during the training stage. The input is convolved by depthwise convolutions of 9 × 9 and 3 × 3, respectively, and then summed. The middle part is the process of RepLKConv reparameterization. First, BN is fused into the weight of the corresponding convolution kernel, and then the two sets of convolution kernels are added according to the center position. The right-hand part is the structure of RepLKConv during the testing stage. The input only needs to be processed by a reparameterized 9 × 9 depthwise convolution. For other sizes of RepLKConv, it needs to change the 9 × 9 part in the illustration to the corresponding size.

which highlights the channels that contain more geometry information and less illumination and color information.

2) Context-Refined Spatial Attention: The procedure of CRSA is shown in Fig. 4. Using the same idea as in CRCA, the information aggregation process in spatial attention is refined. Similarly, for a given input $F \in \mathbb{R}^{C\times H\times W}$, we will generate a set of weights for aggregate information on pixels. F first undergoes average pooling to reduce the size to 2×2, with a total of four sets of vectors. Then, we use 3×3 convolution to linearly transform these four sets of vectors and exchange information within and between vectors so that four sets of weights can selectively focus on different channels to aggregate different spatial features. Then, we normalize each vector using softmax to get the spatial context refinement weight $W \in \mathbb{R}^{C\times 4}$. Afterward, we use multiplying and adding to aggregate the information of the input features to obtain the aggregated spatial features S. The specific process can be formulated as follows:

$$S_t(i, j) = \sum_{k=1}^{C} F_k(i, j) \times W_k(t) \tag{2}$$

where t refers to the index of the four vectors in W. i, j, and k are the indexes in height, width, and channel dimensions, respectively. After getting the spatial features S, the same as SRCA, we use RepLKConv with a kernel size of 17×17 to process the features in order to quickly establish the association in space, and then go through a 1×1 convolution to combine four channels into one channel to get the final spatial attention score S'. Spatial attention can especially highlight the distinct features between different categories. By utilizing contextual information, the semantic spatial features can be precisely learned which results in boundaries between different categories.

3) RepLKConv: For a long time, small convolution kernels are generally used. However, the emergence of the vision transformer has made people see a new possibility. Much work has explained why the vision transformer is powerful from various aspects, including the vast receptive field. However, in a convolutional neural network, simply expanding the convolution kernel does not improve the effect but may lead to a worse effect. Focusing on this problem, Ding et al. [33] proposed a way to effectively use large convolution kernels, which we call RepLKConv here. The details of RepLKConv are shown in Fig. 5. By using small convolution kernels in the residual branch for processing, large convolution kernels can also capture small-scale patterns. Using the reparameterization method can effectively alleviate the optimization issue of large convolution kernels.

With CRCA and CRSA, we need to extract attention scores for features and embed them back into features. Fig. 6 shows the overall process of CRAM. The order of CRCA and CRSA is determined by experiments, which will be mentioned in Section IV. The input F first passes CRCA to obtain the channel attention score C', and then embeds the channel attention into the F by elementwise addition and depthwise convolution. Next, the spatial attention is processed according to the same steps, and the final result O after the attention enhancement is obtained. Different from the common

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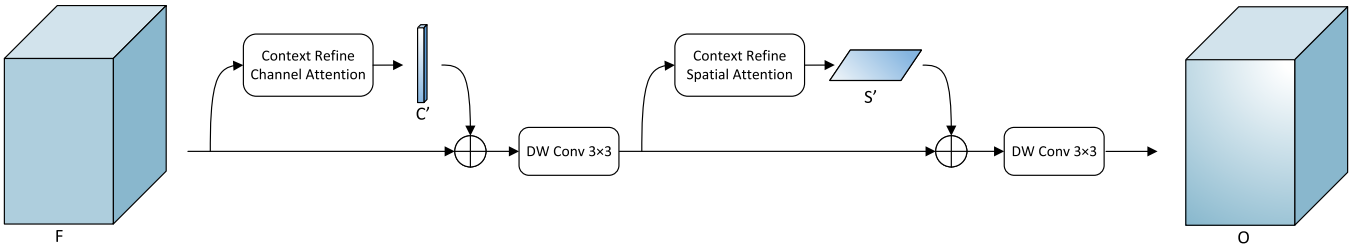


Fig. 6. Overall structure of CRAM. The computed attention scores are then embedded back into the input via elementwise addition and depthwise convolution.

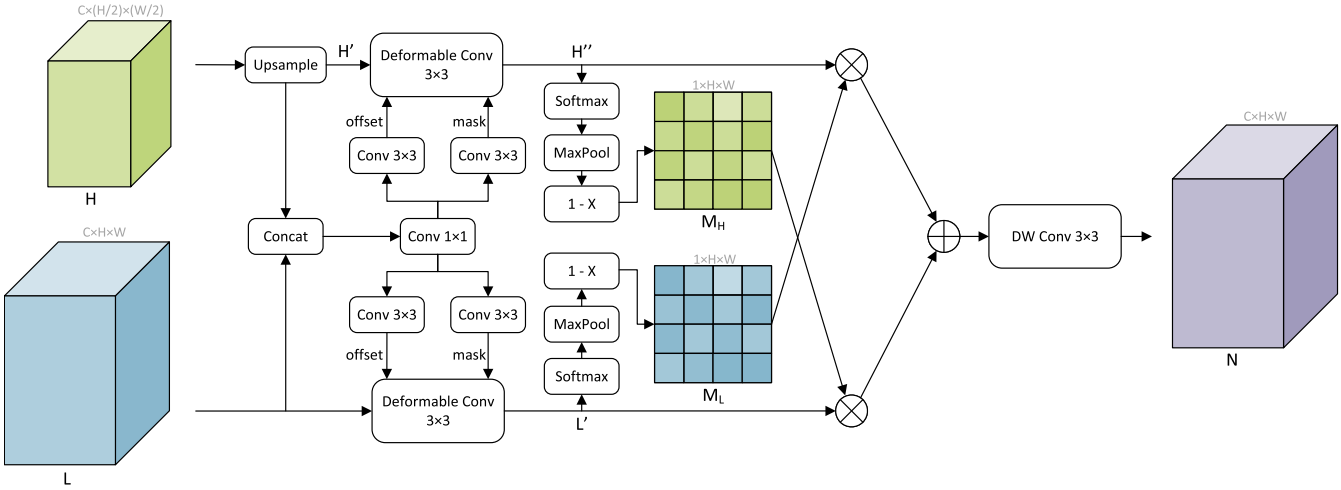


Fig. 7. Structure of ICFFB. Refining the fusion process of adjacent scale features can enable SFN to obtain better fusion results.

elementwise multiplication in the attention mechanism, the reason why we use elementwise addition for combination is mainly to try not to destroy the original features. Through a method similar to residual learning, an attention map is learned and added to the original features as a supplement. The following formula can express the specific details:

$$F' = \theta(F + \mathrm{CRCA}(F)) \tag{3}$$

$$O = \phi\big(F' + \mathrm{CRSA}\big(F'\big)\big) \tag{4}$$

where θ and ϕ both denote the transformation function implemented by depthwise convolution.

C. Stair Fusion Network

In the encoder–decoder structure, we can generally get features of multiple scales from the encoder. The high-level features are small in size and rich in semantic information. Low-level features are large in size and rich in spatial information. Semantic information is critical to classification and spatial information helps to generate accurate edges. Only features that are rich in semantic and spatial information can obtain accurate segmentation results. In order to do this, it becomes essential to fuse multiscale features. In existing works, such as U-Net, MANet, and LANet, there is only one path from top to bottom in feature fusion. Semantic information cannot be sufficiently transferred to low-level features at this time.

In order to get a better feature representation, we propose SFN. The structure is boxed with a dotted line in Fig. 2. We keep the result of the fusion of adjacent scale features and continue to fuse until only one feature remains. The visualized feature map on the right-hand side of Fig. 2 shows that by constructing three top-down feature fusion paths, semantic information and spatial information can be continuously fused.

D. Interaction and Complementary Feature Fusion Block

There are many ways of feature fusion, and the simplest one is to add multiple features element by element or concatenate them together and then generate fused features via convolution. However, using this simple method for processing remote sensing images is unreasonable. The complex information in the features extracted from remote sensing images will be lost due to rough feature fusion. Furthermore, considering the multiscale factor in our proposed network, the multiscale features given by the backbone network are obtained from different stages. Therefore, the features of different scales are different in information abstraction and representation. Considering the traits of the convolutional neural network, the scales of the objects concerned by features of different scales will also be different.

Based on these considerations, we propose ICFFB as a feature fusion block in SFN. Through interaction, the differences in information abstraction and representation between features will be narrowed, and through complementarity, we could focus on objects of all scales as much as possible. Fig. 7 shows the specific structure of ICFFB. For a set of input features, $H \in \mathbb{R}^{C\times(H/2)\times(W/2)}$ is a high-level feature, and $L \in \mathbb{R}^{C\times H\times W}$ is a low-level feature. H will first perform an upsampling to obtain $H' \in \mathbb{R}^{C\times H\times W}$ so that the size is consistent with the low-level features. To achieve the interaction, we use a type of convolution with multiple input parameters, deformable

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convolution [48], [49]. In addition to the input features, the deformable convolution has two other inputs, including offset and mask. Among them, the offset is used to control the offset of each position in the convolution kernel so that the convolution kernel can change its shape to focus on the region of interest, and the mask is used to control the reduction of convolution kernel focus beyond the region of interest. Usually, the offset and mask of the deformable convolution are determined by the input features. However, considering the difference between features of different scales, and in order to make features better integrated, we can interact with each other before the fusion to narrow the difference between them. Therefore, we use features of different scales to generate offset and mask simultaneously. Realize the interaction between features of different scales so that they can be better fused together. Both H' and L are processed using deformable convolution, but the offset and mask of deformable convolution are jointly generated by H' and L. H' and L are first concatenated and fused via a 1×1 convolutional layer. Then, the offset and mask are obtained via a 3×3 convolutional layer. After deformable convolution processing, we get H'' and L'. Specifically, this process can be expressed by the following formula:

$$I = \varphi\big(\text{Concat}\big[H', L\big]\big) \tag{5}$$

$$H'' = \text{DCN}\big(H', \beta(I), \gamma(I)\big) \tag{6}$$

$$L' = \text{DCN}(L, \eta(I), \mu(I)) \tag{7}$$

where DCN refers to deformable convolution, Concat refers to concatenate operation, φ denotes the transformation function implemented by 1×1 convolution, and β, γ, η, and μ denote the different transformation functions implemented by 3×3 convolution. Then, we perform softmax normalization on each pixel and find the maximum value. In the map obtained at this time, we can consider that the positions with large values are well represented in the original feature. Therefore, after subtracting from 1, we get $M_H \in \mathbb{R}^{1\times H\times W}$ and $M_L \in \mathbb{R}^{1\times H\times W}$, and the meaning is opposite to original features. This process can be expressed as the following formula, for the pixel in ith row and jth column:

$$M_H(i, j) = 1 - \text{Max}\big(\text{Softmax}\big(H''(i, j)\big)\big) \tag{8}$$

$$M_L(i, j) = 1 - \text{Max}\big(\text{Softmax}\big(L'(i, j)\big)\big) \tag{9}$$

where i and j are the indexes in height and width dimensions, respectively. Subsequent elementwise multiplication of H'' and L' with M_L and M_H, respectively, strengthens the weakly represented position in the other features, achieving complementarity. Finally, we use elementwise add and depthwise convolution to obtain the fused feature N. As shown in Fig. 2, features in lower layers contain spatial detail information. With the SFN and proposed ICFFB, such information can be sufficiently and progressively integrated into the fused features. Then, there are obvious boundaries between different categories in the fused features.

IV. Experiments

This section will conduct some experiments to test the proposed method. First, we evaluate the proposed modules using ablation experiments. Next, we compare the entire model with some recent state-of-the-art methods. Finally, we conduct some visualization experiments to analyze the operating principles of some modules qualitatively.

A. Datasets

1) ISPRS Vaihingen Dataset: The Vaihingen dataset under consideration comprises 33 patches of varying sizes, with an average resolution of 2494 × 2064 pixels. The true orthophoto (TOP) and the DSM have a ground sampling distance (GSD) of 9 cm. The TOP comprises three bands corresponding to near-infrared (IR), red (R), and green (G) bands obtained from the camera. The annotation encompasses six categories: impervious surfaces, buildings, low vegetation, trees, cars, and clusters. For training, 16 images have been selected, namely, image ID: 1, 3, 5, 7, 11, 13, 15, 17, 21, 23, 26, 28, 30, 32, 34, and 37, with the remaining 17 images being reserved for testing.

2) ISPRS Potsdam Dataset: The Potsdam dataset under consideration comprises 38 patches of identical sizes with a 6000 × 6000 pixels resolution. The TOP and the DSM have a GSD of 5 cm. Each TOP has three combinations of bands, namely, IR-R-G, R-G-B, and R-G-B-IR. The labeled categories are the same as in the Vaihingen dataset. TOP with R-G-B and DSM are utilized as input in the experiment, while TOP with IR-R-G and TOP with R-G-B-IR are not involved. For training, 24 images have been selected, namely, image ID: 2_10, 2_11, 2_12, 3_10, 3_11, 3_12, 4_10, 4_11, 4_12, 5_10, 5_11, 5_12, 6_7, 6_8, 6_9, 6_10, 6_11, 6_12, 7_7, 7_8, 7_9, 7_10, 7_11, and 7_12, with the remaining 14 images being reserved for testing.

3) LoveDA Dataset: The LoveDA dataset [50] contains 5987 remote sensing images, each with a size of 1024 × 1024 pixels. The image has a GSD of 0.3 m. The dataset contains seven categories: background, building, road, water, barren, forest, and agriculture. Among them, 2522 images are used for training, 1669 images are used for validation, and 1796 images are used for testing. The dataset contains both urban and rural images.

B. Evaluation Metrics

We use the mean intersection over union (mIoU) and mean F1 (mF1) score to evaluate the performance of all methods. Both are calculated from the confusion matrix, including true positive (TP), false positive (FP), true negative (TN), and false negative (FN). The meaning of IoU is the ratio of the intersecting part to the union part, which is calculated using the following formula:

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}}. \tag{10}$$

The formula for calculating the F1 score is as follows:

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{11}$$

$$\text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{12}$$

$$\text{F1} = 2 \times \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}}. \tag{13}$$

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After calculating each category's IoU and F1 scores through the above-mentioned formula, they are averaged to obtain the mIoU and mF1 scores.

C. Implementation Details

We use ResNet-18 as the backbone with pretrained weights on ImageNet. We utilize AdamW [51] as the optimizer. The learning rate is set to 0.0006, the betas are set to 0.9 and 0.999, and the weight decay is set to 0.01. We adopt the poly learning rate strategy, where the parameter power is set to 1.2. We employ the cross-entropy loss function to calculate the loss for semantic segmentation. For the two ISPRS datasets, we first crop all TOPs into 1024 × 1024 pixel patches and finally randomly crop a 512 × 512 pixel patch from them as input. In all experiments, only G2GNet used the DSM. For the LoveDA dataset, we randomly crop a 512 × 512 pixel patch as input. During training, vertical and horizontal flips, random rotations, and photometric distortion are used for data augmentation. We use vertical and horizontal flips as test-time data augmentation during testing. All experiments are performed on a single NVIDIA Geforce 2080Ti.

D. Ablation Study

We conduct ablation experiments on two datasets to demonstrate the proposed modules' efficacy. Our baseline model is a simple U-shaped network with ResNet-18 as the backbone. Considering that under limited memory, the backbone network and the encoder are relatively balanced in the number of parameters and calculations, the network's overall performance will be better. Moreover, considering the lightweight, we finally chose ResNet-18 as the backbone network. The specific experimental results are shown in Table I, where SFN_struct refers to the ablation experiment for the validity of the SFN structure and only a simple feature fusion method is used instead of ICFFB.

On the Vaihingen dataset, incorporating our proposed modules brings significant improvements compared with the baseline models. Adding CRAM to the baseline can improve the model by 2.08% on mIoU and 1.30% on mF1 score. In order to verify the effectiveness of the SFN structure itself, we add the SFN feature fusion structure to the baseline, and the model performance improved by 1.30% on mIoU and 0.83% on mF1. At the same time, we add ICFFB to SFN to make it into a complete state, and mIoU and mF1 increased by 1.74% and 1.06%, respectively. Finally, we fuse all the modules, achieving 83.86% on mIoU and 91.09% on mF1.

On the Potsdam dataset, our proposed modules also bring performance improvements, but the improvement is relatively small compared with the results on the Vaihingen dataset. Adding CRAM to the baseline improves the model by 0.84% and 0.50% on mIoU and mF1, respectively. Adding the SFN structure to the baseline improves 0.47%mIoU and 0.27%mF1 compared with the baseline model. After adding ICFFB to make SFN a complete form, mIoU is further increased by 0.76%, and mF1 is also increased by 0.44%. Finally, all modules are added, and our method achieves 87.06% on mIoU and 92.97% on mF1. The two metrics are improved by 1.66% and 0.99% compared with the baseline model.

TABLE I
RESULTS OF ABLATION EXPERIMENTS FOR THE PROPOSED METHOD

Dataset	Method	mIoU	mF1
Vaihingen	Base	80.03	88.72
	Base + CRAM	82.11	90.02
	Base + SFN_struct	81.33	89.55
	Base + SFN	83.07	90.61
	Base + CRAM + SFN	83.86	91.09
Potsdam	Base	85.40	91.98
	Base + CRAM	86.24	92.48
	Base + SFN_struct	85.87	92.25
	Base + SFN	86.63	92.69
	Base + CRAM + SFN	87.06	92.97

E. Comparison With the State-of-the-Art Methods

We compare the proposed method with recent state-of-the-art methods, including FCN [7], PSPNet [11], DeepLabV3+ [35], DANet [25], UPerNet [30], dense dilated convolutions merging network (DDCM-Net) [52], LANet [19], MANet [26], CF-Net [20], G2GNet [53], and UNetFormer [18]. Among them, the first five methods are mainly used for the semantic segmentation of natural images, and the remaining methods are biased toward processing remote sensing images.

In the selection of the backbone, we try to choose backbones with similar performance and close to the backbone used in the original paper of these methods. Specifically, include the following three, VGG-16 [54], ResNet-18 [47], and Swin-T [55]. The detailed correspondence will be reflected in the table of the experimental results.

1) Results on the Vaihingen Dataset: Table II demonstrates the quantitative results of all methods on the Vaihingen dataset. The middle part of Table II is the IoU of each category, and the right side is the overall evaluation metrics. By comparing the metrics of mIoU and mF1, we can know that SFCRNet is better than others. PSPNet and DeepLabV3+ use pooling and dilation convolution to obtain more comprehensive context information and get good results. DANet improves the global modeling ability by using global pixel attention and channel attention, which has a 1.04% improvement in mIoU compared with the above-mentioned two methods. MANet mathematically simplifies global attention and uses multiscale information from the encoder. UPerNet uses the Swin transformer as the backbone and pays more attention to global information in the encoding stage. After comparison, it can be found that UPerNet performs superiorly in the category of impervious surfaces with a larger area, reflecting the powerful global capabilities of the Swin transformer. DDCM-Net increases the receptive field exponentially through dense dilation convolution and has a 0.34% improvement on mIoU compared with DANet. LANet uses attention to make high-level features guide low-level features, bridge the gap between the two, and get better results. CF-Net incorporates a variety of segmentation network framework structures to achieve better performance. G2GNet achieves 82.39% mIoU by gathering information from other modalities and using it to guide RGB image features. UNetFormer combines convolution

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TABLE II
EXPERIMENTAL RESULTS ON THE ISPRS VAIHINGEN DATASET

Method	Backbone	Impervious surface	Building	Low vegetation	Tree	Car	mIoU(%)	mF1(%)
FCN [7]	VGG-16	82.18	89.30	69.87	79.43	43.13	72.78	83.13
PSPNet [11]	ResNet-18	84.51	90.37	70.27	80.55	74.06	79.95	88.68
DeepLabV3+ [35]	ResNet-18	84.96	90.64	71.35	81.62	73.33	80.38	88.95
DANet [25]	ResNet-18	85.58	91.66	71.58	80.97	77.30	81.42	89.60
UPerNet [30]	Swin-T	87.22	92.29	73.16	81.26	77.14	82.22	90.08
DDCM-Net [52]	ResNet-18	86.06	91.60	72.64	81.65	76.82	81.76	89.81
LANet [19]	ResNet-18	85.61	91.97	72.23	81.27	80.85	82.39	90.20
MANet [26]	ResNet-18	86.65	91.91	72.27	81.45	77.67	81.99	89.95
CF-Net [20]	VGG-16	86.60	91.18	72.59	81.50	79.39	82.25	90.13
G2GNet [53]	ResNet-18	86.12	91.40	72.29	81.36	80.80	82.39	90.21
UNetFormer [18]	ResNet-18	86.96	92.07	73.30	81.78	78.87	82.60	90.33
SFCRNet	ResNet-18	**87.85**	**92.72**	**74.21**	**82.12**	**82.38**	**83.86**	**91.09**

TABLE III
EXPERIMENTAL RESULTS ON THE ISPRS POTSDAM DATASET

Method	Backbone	Impervious surface	Building	Low vegetation	Tree	Car	mIoU(%)	mF1(%)
FCN [7]	VGG-16	84.06	91.37	75.66	78.51	78.54	81.63	89.78
PSPNet [11]	ResNet-18	86.20	92.43	76.20	79.12	91.90	85.17	91.85
DeepLabV3+ [35]	ResNet-18	87.01	93.60	77.22	79.26	91.28	85.67	92.15
DANet [25]	ResNet-18	87.13	93.52	77.45	79.82	**92.32**	86.05	92.37
UPerNet [30]	Swin-T	87.74	93.59	77.91	79.78	90.45	85.89	92.30
DDCM-Net [52]	ResNet-18	87.17	93.36	77.10	79.57	91.38	85.72	92.18
LANet [19]	ResNet-18	87.02	93.27	75.33	80.12	92.18	85.58	92.08
MANet [26]	ResNet-18	88.11	94.12	77.91	79.96	91.71	86.36	92.55
CF-Net [20]	VGG-16	87.32	93.34	77.23	79.26	91.87	85.80	92.23
G2GNet [53]	ResNet-18	87.06	93.20	77.52	79.67	92.18	85.93	92.30
UNetFormer [18]	ResNet-18	88.00	94.13	77.58	79.66	91.86	86.24	92.48
SFCRNet	ResNet-18	**88.49**	**94.71**	**78.96**	**80.93**	92.21	**87.06**	**92.97**

and transformer to preserve spatial details while improving global capabilities, gaining 82.60% mIoU.

The visualized experimental results are shown in Fig. 8. As can be seen from Fig. 8, SFCRNet has a good performance on the Vaihingen dataset. By refining the process of information aggregation in attention and a suitable feature fusion method, it can still perform well in complex semantic environments. The first two images are with severe shadows which significantly influence the segmentation accuracy. As shown in the surrounded regions, many existing methods wrongly segment the shaded regions, while SFCRNet has better semantic extraction capability and can generate more accurate boundaries. In the third image, the building is highly similar to the impervious surface. Other methods have a certain degree of misjudgment, with the refined contextual information and multiscale feature fusion, SRCFNet is able to accurately deal with the similar features between categories and generate accurate boundaries.

2) Results on the Potsdam Dataset: Table III shows the quantitative experimental results of all methods on the Potsdam dataset. SFCRNet again achieves the best results on both metrics. Since the Potsdam dataset has more abundant training data, the global attention modules in DANet and MANet have been more fully trained, which is much better than the performance on the Vaihingen dataset. UNetFormer, which also uses global attention, performs well on both datasets due to the combination of convolutions.

Similarly, we also visualized the experimental results on the Potsdam dataset, as shown in Fig. 9. Due to the adequate fusion of multiscale features by SFN, the features obtained by SFCRNet retain sufficient spatial details while having semantic information, making it clear to segment the trail on the right-hand side of the lawn in the first image. DDCM-Net quickly expands the receptive field, making it perform well on the overall segmentation but lacking details restoration. LANet lacks the use of intermediate features given by the backbone network, and the edge of the segmented area is accurate, but the internal details of the area still need to be included. On the right-hand side of the second image, the building is very similar to the texture of the impervious surface. Other methods have more or less classification errors, classifying buildings as impervious surfaces or vice versa. SFCRNet accurately classifies the two through the association of contextual information and has better edge performance. In the shaded area in the middle of the third image, most methods did not accurately classify that it belongs to the building, LANet roughly segmented it, G2GNet incorporates DSM information and also segments the region to some extent, SFCRNet is able to more accurately distinguish similar categories from shadows and produce more accurate edges by virtue of the refined contextual information. Moreover, many

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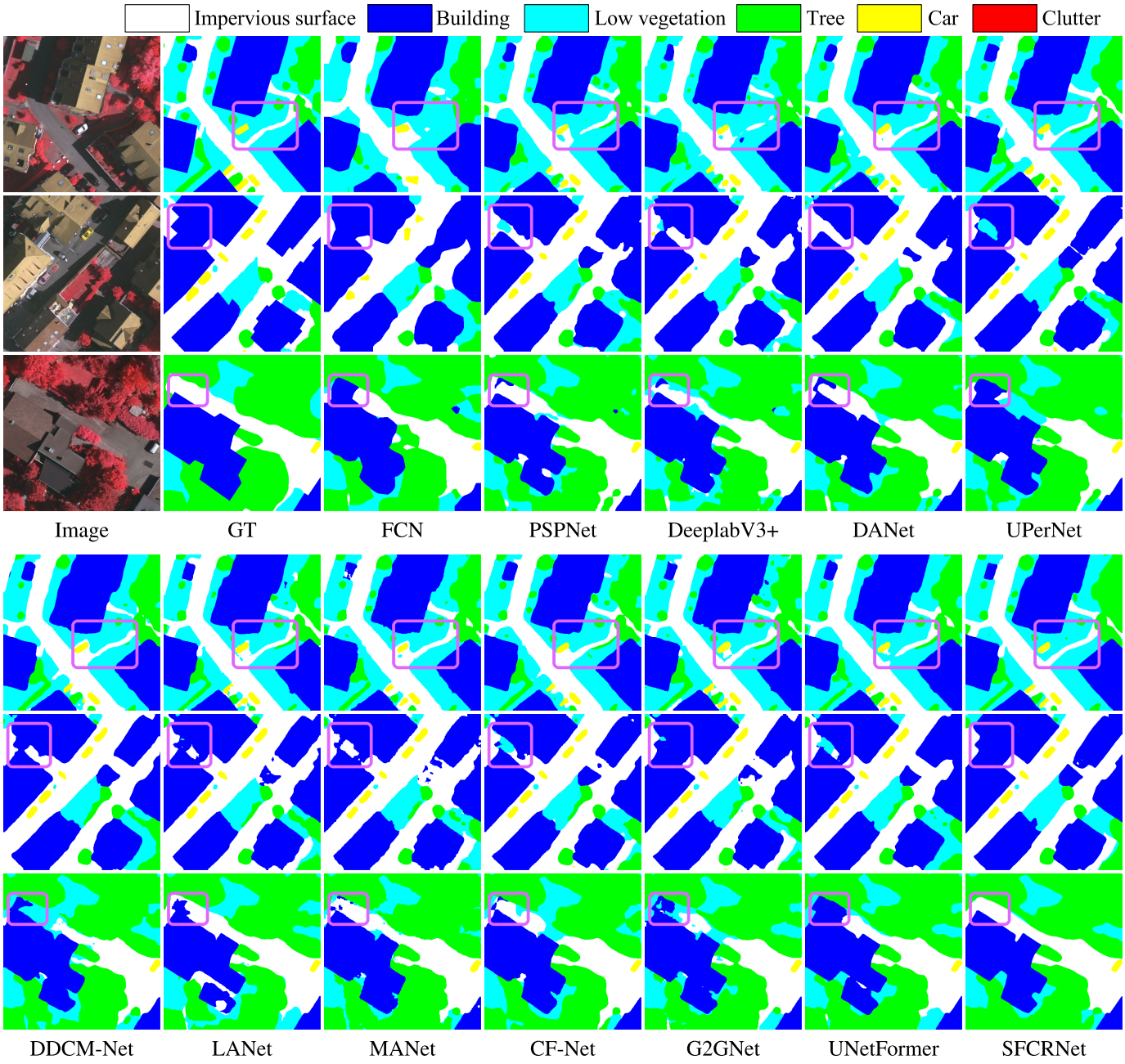


Fig. 8. Visualization of experimental results on the ISPRS Vaihingen Dataset.

methods have the problem of broken area boundaries at the edge of the shadow.

3) Results on the LoveDA Dataset: Since there is no data on other modalities similar to DSM in the LoveDA dataset, G2GNet does not participate in the comparison. Table IV shows the quantitative experimental results of each method on the LoveDA dataset. Since the official provides the calculation of the test set score and only the IoU score is provided, there is no mF1 score in Table IV. SFCRNet achieves the highest score on mIoU and also the highest on IoU for each category. The method using VGG-16 and Swin-T as the backbone network performs relatively better on the LoveDA dataset, indicating that this dataset has higher requirements for the feature extraction capabilities of the backbone network. Moreover, because the dataset is dominated by large landcover categories and with stronger feature extraction capabilities of VGG-16, FCN surpassed PSPNet and DeepLabV3+. However, SFCRNet still achieved the highest score when the backbone network was ResNet-18, reflecting its powerful decoding ability.

The visualization results on the LoveDA validation set are shown in Fig. 10. In general, SFCRNet has an advantage in target capture and edge accuracy. For buildings, SFCR-Net can be segmented just right without over-enlarging or over-shrinking. For barren, SFCRNet has a stronger ability to judge and performs better than other methods in the second and third images. The use of CRSA can refine the detail information such as margins. Incorporated with the stair fusion and proposed fusion module, SFCRNet generates accurate segmentation without over-enlarging or over-shrinking.

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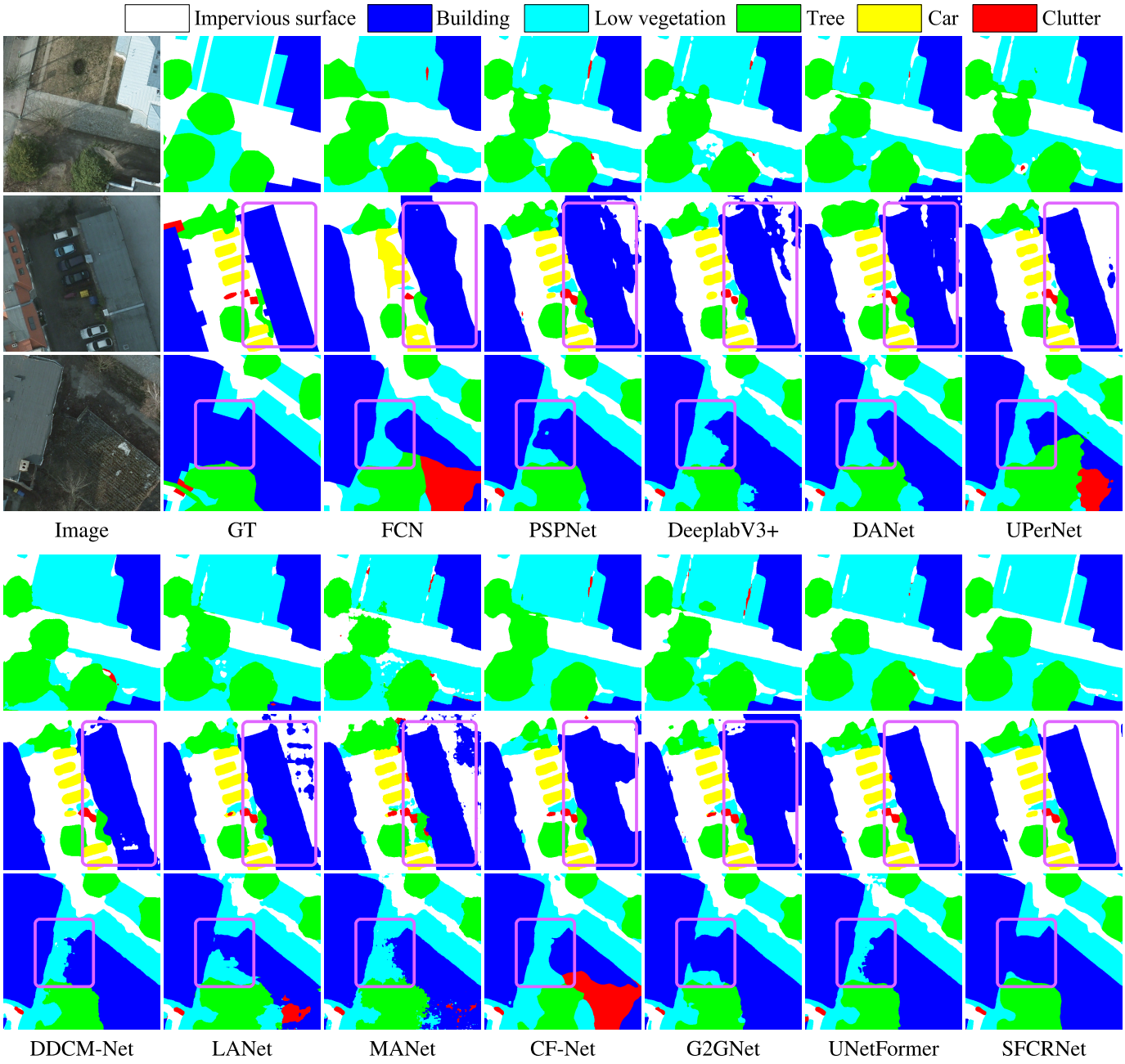


Fig. 9. Visualization of experimental results on the ISPRS Potsdam Dataset.

TABLE IV
EXPERIMENTAL RESULTS ON THE LOVEDA DATASET

Method	Backbone	Background	Building	Road	Water	Barren	Forest	Agricultural	mIoU(%)
FCN [7]	VGG-16	43.21	53.01	53.05	75.48	17.69	44.47	57.72	49.23
PSPNet [11]	ResNet-18	44.20	51.73	51.75	76.01	10.70	43.75	58.57	48.10
DeepLabV3+ [35]	ResNet-18	40.76	54.90	54.56	73.85	15.14	45.56	57.55	48.90
DANet [25]	ResNet-18	43.47	55.35	54.64	76.47	19.27	45.18	57.46	50.26
UPerNet [30]	Swin-T	45.42	55.50	59.20	79.54	18.63	46.13	59.08	51.93
DDCM-Net [52]	ResNet-18	42.11	52.65	54.88	76.95	15.42	45.60	58.58	49.46
LANet [19]	ResNet-18	43.96	51.89	58.31	79.15	18.96	45.03	57.54	50.69
MANet [26]	ResNet-18	44.36	54.44	56.48	80.59	17.75	44.21	58.57	50.91
CF-Net [20]	VGG-16	44.67	55.50	57.87	79.32	17.65	44.34	62.45	51.68
UNetFormer [18]	ResNet-18	44.48	55.67	60.19	80.27	18.01	45.62	59.21	51.92
SFCRNet	ResNet-18	**46.78**	**58.28**	**61.73**	**81.22**	**19.89**	**46.71**	**60.11**	**53.53**

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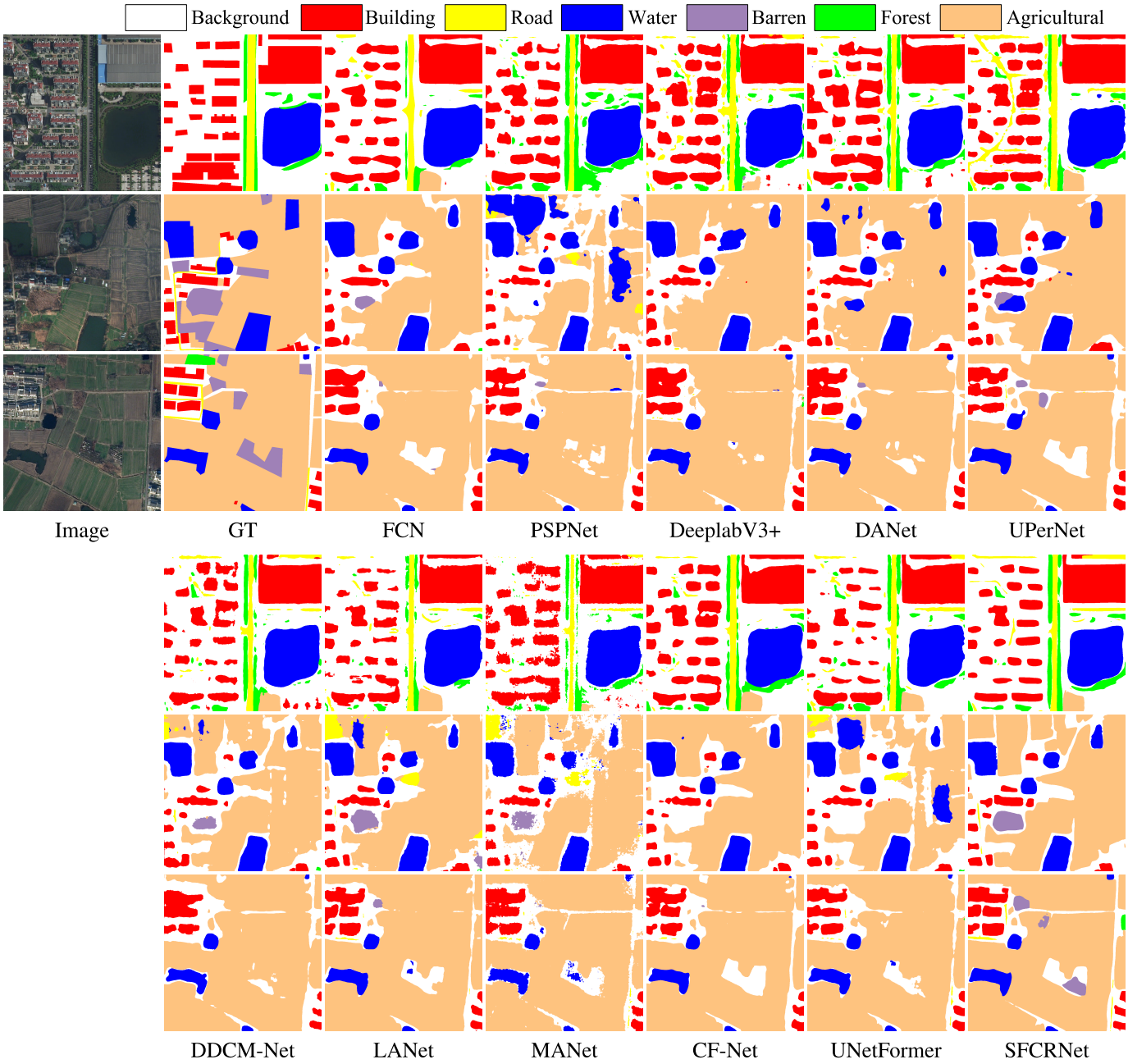


Fig. 10. Visualization of experimental results on the LoveDA Dataset.

F. Experiment on CRAM

1) Experiments With Order of Attention in CRAM: To determine the order of channel and spatial attention in CRAM, we conduct experiments on the Vaihingen dataset. We tried three combinations: channel first and then spatial, spatial first and then channel, and channel and spatial in parallel. We added these three combinations to the baseline, and the specific experimental results are shown in Table V. It can be seen from Table V that the order of channel first and then spatial is optimal on both mIoU and mF1. Therefore, we finally decided to determine the order of the two attentions in CRAM as first channel attention and then spatial attention.

TABLE V
RESULTS OF EXPERIMENTS FOR DIFFERENT ORDER IN CRAM ON THE VAIHINGEN DATASET

Method	mIoU(%)	mF1(%)
CRAM-channel-spatial	**82.11**	**90.02**
CRAM-spatial-channel	82.03	89.98
CRAM-parallel	81.88	89.89

2) Effect of RepLKConv in CRAM: In order to verify the effectiveness of RepLKConv and determine the final kernel size of RepLKConv, we conducted experiments. Because RepLKConv is made based on depthwise convolution in the experiment, we used three different convolutions in CRAM: standard convolution, depthwise convolution, and RepLKConv. At the same time, we also compared the impact of different kernel sizes on the segmentation results. The specific experimental results are shown in Table VI, where

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TABLE VI
RESULTS OF DIFFERENT CONVOLUTION TYPES AND DIFFERENT CONVOLUTION KERNEL SIZES ON THE VAIHINGEN DATASET

Method	mIoU(%)	mF1(%)	Params	FLOPs
Conv 3×3	83.49	90.88	168.37K	10.76G
Conv 17×17	83.14	90.66	4.76M	311.41G
DW Conv 3×3	83.55	90.91	22.04K	1.17G
DW Conv 17×17	83.29	90.75	57.90K	3.52G
RepLKConv 5×5	83.65	90.97	24.11K	1.31G
RepLKConv 9×9	83.70	91.00	31.28K	1.78G
RepLKConv 13×13	83.62	90.95	42.54K	2.52G
RepLKConv 17×17	**83.86**	**91.09**	57.90K	3.52G
RepLKConv 21×21	83.67	90.99	77.36K	4.80G
RepLKConv 25×25	83.81	91.07	100.91K	6.34G

TABLE VII
RESULTS OF DIFFERENT ATTENTION MODULES ON THE VAIHINGEN DATASET

Method	mIoU(%)	mF1(%)	Params	FLOPs
SE [21]	81.60	89.71	11.77M	56.77G
ECA [32]	81.33	89.54	11.76M	56.77G
CBAM [22]	81.51	89.65	11.77M	56.79G
SA [56]	81.42	89.59	11.76M	56.78G
SK [57]	81.78	89.82	14.04M	105.34G
CRAM-mul	81.43	89.61	12.76M	61.55G
CRAM	**82.11**	**90.02**	12.76M	61.79G

flops are calculated when the input channel is 128 and the input size is 256 × 256. Standard convolution is too computationally expensive in comparison, and the effect is relatively general. Moreover, the score decreased after expanding the convolution kernel. We guessed that it was difficult to optimize the convolution kernel that was too large. Depthwise convolution looks slightly better but still encounters the problem of score drop when expanding the convolution kernel. RepLKConv is different. Due to the existence of a 3 × 3 auxiliary branch, the score shows an overall upward trend as the kernel increases. However, after comprehensively considering the calculation amount and the segmentation result, we decided to use 17 × 17 RepLKConv.

3) Compare With Other Attention Modules: In order to verify the superiority of multiplication and addition in CRAM for information aggregation, we compared some attention mechanisms, including SE [21], efficient channel attention (ECA) [32], CBAM [22], shuffle attention (SA) [56], and selective kernel (SK) [57]. In order to compare the performance of elementwise addition and elementwise multiplication in CRAM, we added elementwise multiplication CRAM to the experiment, denoted as CRAM-mul. The information aggregation part of these attention mechanisms uses a simple mean or maximum value. The specific experimental results are shown in Table VII. It can be seen that CRAM has the highest mIoU and mF1 scores and has a good improvement compared with other attention mechanisms. This demonstrates the effectiveness of our proposed information aggregation method, and the introduction of CRAM will not significantly increase the number of parameters and computational complexity of the model.

TABLE VIII
RESULTS OF EXPERIMENTS FOR WITH OR WITHOUT DEFORMABLE CONVOLUTION ON THE VAIHINGEN DATASET

Method	Deformable Convolution	mIoU(%)	mF1(%)
Base + SFN		82.17	90.06
Base + SFN	✓	**83.07**	**90.61**

4) Visualization of CRSA: To further explore why SFCRNet performs well and why CRAM works. We visualized the attention score map obtained by CRSA and compared it with the input image and the ground truth. Since spatial attention mainly works on low-level features with rich spatial information, we visualize the attention score maps obtained from 1/4 downsampled features. We visualized both the Vaihingen and Potsdam datasets, as shown in Fig. 11. The second column is the result of the visualization of the spatial attention score map. The darker the color, the smaller the value; the brighter the color, the larger the value.

Fig. 11 shows that CRSA mainly focuses on categories with large connected areas, such as impervious surfaces, low vegetation, and trees. Less attention is paid to categories with small individual regions, such as buildings and cars. Generally, the receptive field of low-level features is relatively small, and the spatial information is relatively prosperous, which means it already has a considerable degree of semantic extraction ability for smaller scale objects. However, for large-scale categories, it is difficult to judge them as a whole in the low-level features and prone to breaks in the middle. CRSA improves the continuity of these categories in low-level features by broadly focusing on large-scale categories. Moreover, after further observation, we found that CRSA will pay more attention to the shaded area, which is especially obvious in the Vaihingen dataset and explains to some extent why SFCRNet performs better in the shaded area.

At the same time, we also notice that the images visualized from CRSA are very different on the two datasets. Through observing and comparing the original images and ground truth in the datasets, we found that impervious surfaces in the Vaihingen dataset have many large connected areas, but low vegetation and trees basically only have small independent regions. In the Potsdam data, many large connected areas exist in these three categories. Based on these observations and the speculation on the working principle of CRSA earlier, the differences in the datasets may indeed cause the trained models to pay attention to the different categories.

We also visualize parts of the spatial attention map in CBAM for comparison. The specific results are shown in the Fig. 11. Fig. 11 shows that CBAM has a more nuanced and messier focus on space. In 1/4 of the attention map, there seems to be no specific attention to certain large areas and no special care for shadowed areas. In the 1/16 and 1/32 attention maps, the attention to shadows is reflected to a certain extent. However, the boundaries are blurred, and the attention is not strong enough.

G. Experiments on Deformable Convolution in ICFFB

In order to further verify the rationality of using deformable convolution in ICFFB, we designed experiments for

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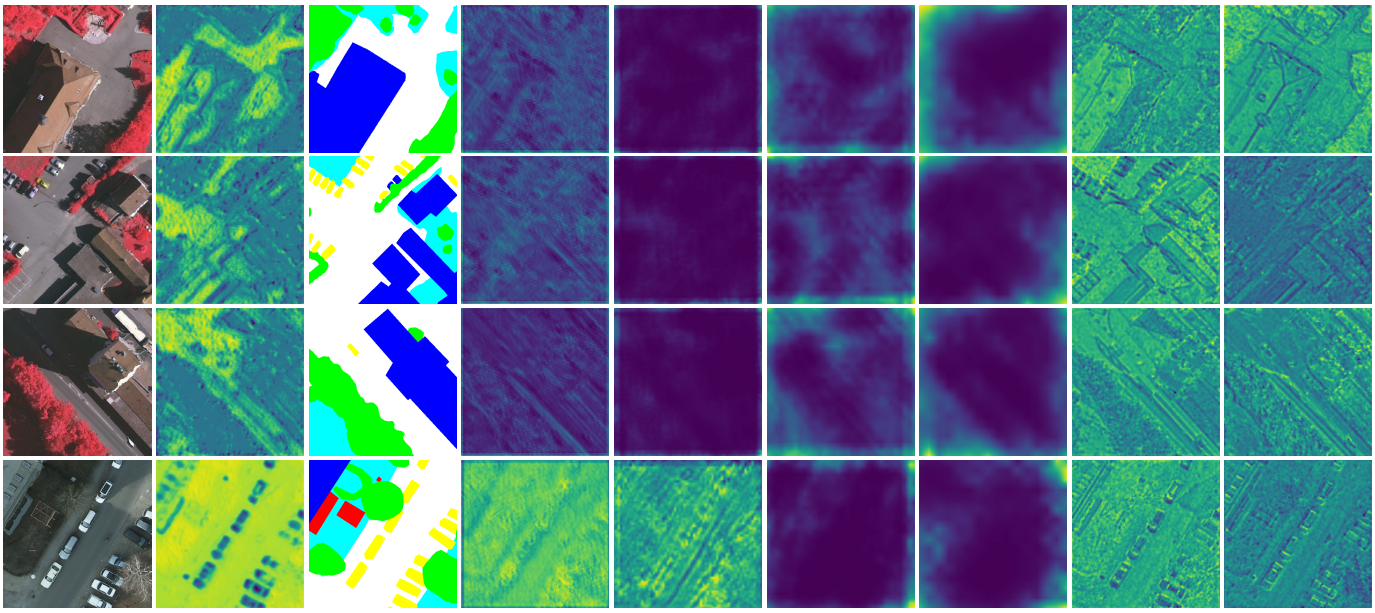


Fig. 11. Visualization of CRSA, CBAM, and features on two ISPRS Datasets. (a) Input image. (b) CRSA attention map obtained from 1/4 downsampled features. (c) Ground truth. (d) CBAM spatial attention map obtained from 1/4 downsampled features. (e) CBAM spatial attention map obtained from 1/8 downsampled features. (f) CBAM spatial attention map obtained from 1/16 downsampled features. (g) CBAM spatial attention map obtained from 1/32 downsampled features. (h) Features rich in shadow information. (i) Features that lack shadow information.

TABLE IX
RESULTS FOR EFFICIENCY ANALYSIS

Method	Backbone	Params	FLOPs	FPS	mIoU(%)
FCN [7]	VGG-16	15.31M	322.01G	26.82	72.78
PSPNet [11]	ResNet-18	12.64M	44.20G	30.74	79.75
DANet [25]	ResNet-18	12.71M	44.54G	20.80	81.42
UPerNet [30]	Swin-T	30.51M	149.72G	19.37	82.22
LANet [19]	ResNet-18	11.45M	43.96G	30.56	82.39
MANet [26]	ResNet-18	12.66M	102.36G	30.95	81.99
CF-Net [20]	VGG-16	17.09M	341.04G	23.63	82.25
SFCRNet	ResNet-18	15.26M	140.58G	13.39	83.86

comparison. We chose Base + SFN in the ablation experiment as the benchmark that used the deformable convolution and the model that replaced the deformable convolution in ICFFB with the standard convolution for comparison. The specific experimental results are shown in Table VIII, and it can be seen that there is a 0.9% improvement in mIoU after using the deformable convolution. The experimental results fully demonstrate that the use of deformable convolution in ICFFB can improve the model.

H. Efficiency Analysis

Computational efficiency is critical to the implementation of the framework. Therefore, we evaluated the model's performance through theoretical estimation and experimental testing and compared it with some other methods. The specific results are shown in Table IX. The parameter quantities are obtained statistically from the model. The calculation amount and frame per second (FPS) are obtained in the case of a three-channel input image with a size of 1024 × 1024. The calculation amount is an estimated theoretical value and is for reference only. FPS is the result of the experimental calculation. mIoU is the result of the Vaihingen dataset. It can be seen that when SFCRNet obtains the highest mIoU, the FPS is relatively low. However, at the same time, the amount of theoretical calculation is not very high. Compared with the two methods using VGG-16 as the backbone network, SFCRNet has a lower FPS but with less calculation. There are two reasons: the introduction of deformable convolution and large kernel DW convolution. In the past few years of rapid development of deep learning, maybe since people rarely use large kernel convolutions, the existing off-the-shelf deep learning tools (such as PyTorch) lack support for large kernel DW convolutions [33]. This leads to a much slower DW convolution inference speed for large kernels. The deformable convolution is due to its late proposal and particularity compared with the standard convolution. There is currently no support from acceleration libraries, such as cuDNN, so the inference speed will be much slower than standard convolution. Therefore, SFCRNet achieves low FPS results in experiments. However, if the optimization of the above-mentioned two points can be completed, the inference efficiency of SFCRNet will be significantly improved.

V. CONCLUSION

In this article, we propose a semantic segmentation framework SFCRNet for remote sensing images based on attention mechanism and feature fusion. CRAM rethinks the information aggregation process in the attention mechanism and uses context-assisted refinement to make the original data more representative during attention processing. In addition, to better fuse multiscale features, we propose SFN and the corresponding fusion module ICFFB through multiple top-down information transfers by fully considering the differences between features of different scales. Experimental results on

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the ISPRS Vaihingen Dataset, the ISPRS Potsdam dataset, and the LoveDA dataset demonstrate the effectiveness of our proposed SFCRNet. From the visualized results, SFCRNet is robust to shadows and has a larger capability in dealing with similar categories. Moreover, with the contextual refined attention and stair fusion strategy, SFCRNet generates more accurate boundaries. However, in terms of efficiency, our method still needs to be improved, especially since the introduction of deformable convolution leads to a slower inference speed of the model. In the future, we will work on the model's compression and improve the model's inference speed. We will further explore semantic segmentation of remote sensing images under unsupervised and weakly supervised conditions to reduce the dependence on the number of labeled samples. Through meta-learning, transfer learning, and other methods, we will transfer the prior knowledge learned from large-scale labeled natural image data to remote sensing images with less data volume.

REFERENCES

[1] S. Jadon, "A survey of loss functions for semantic segmentation," in *Proc. IEEE Conf. Comput. Intell. Bioinf. Comput. Biol. (CIBCB)*, Oct. 2020, pp. 1–7.

[2] H. M. Rajesh, "Application of remote sensing and GIS in mineral resource mapping—An overview," *J. Mineralogical Petrological Sci.*, vol. 99, no. 3, pp. 83–103, 2004.

[3] W. Zang, J. Lin, Y. Wang, and H. Tao, "Investigating small-scale water pollution with UAV remote sensing technology," in *Proc. World Autom. Congr.*, Jun. 2012, pp. 1–4.

[4] J. Zhao, H. Ghedira, and M. Temimi, "Detection of oil pollution in the Arabian Gulf using optical remote sensing imagery," in *Proc. IEEE Geosci. Remote Sens. Symp.*, Jul. 2014, pp. 1453–1456.

[5] Q. Zhang and K. C. Seto, "Mapping urbanization dynamics at regional and global scales using multi-temporal DMSP/OLS nighttime light data," *Remote Sens. Environ.*, vol. 115, no. 9, pp. 2320–2329, Sep. 2011.

[6] D. Marcos, M. Volpi, B. Kellenberger, and D. Tuia, "Land cover mapping at very high resolution with rotation equivariant CNNs: Towards small yet accurate models," *ISPRS J. Photogramm. Remote Sens.*, vol. 145, pp. 96–107, Nov. 2018.

[7] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.

[8] V. Badrinarayanan, A. Kendall, and R. Cipolla, "SegNet: A deep convolutional encoder–decoder architecture for image segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 12, pp. 2481–2495, Dec. 2017.

[9] C. Yu, J. Wang, C. Peng, C. Gao, G. Yu, and N. Sang, "Learning a discriminative feature network for semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 1857–1866.

[10] X. Li, Z. Zhong, J. Wu, Y. Yang, Z. Lin, and H. Liu, "Expectation-maximization attention networks for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9167–9176.

[11] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2881–2890.

[12] L.-C. Chen, G. Papandreou, F. Schroff, and H. Adam, "Rethinking atrous convolution for semantic image segmentation," 2017, *arXiv:1706.05587*.

[13] J. Zhao et al., "Multi-source collaborative enhanced for remote sensing images semantic segmentation," *Neurocomputing*, vol. 493, pp. 76–90, Jul. 2022.

[14] L. Mou, Y. Hua, and X. X. Zhu, "Relation matters: Relational context-aware fully convolutional network for semantic segmentation of high-resolution aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 11, pp. 7557–7569, Nov. 2020.

[15] S. Sun, L. Mu, L. Wang, P. Liu, X. Liu, and Y. Zhang, "Semantic segmentation for buildings of large intra-class variation in remote sensing images with O-GAN," *Remote Sens.*, vol. 13, no. 3, p. 475, Jan. 2021.

[16] S. Liu, W. Ding, C. Liu, Y. Liu, Y. Wang, and H. Li, "ERN: Edge loss reinforced semantic segmentation network for remote sensing images," *Remote Sens.*, vol. 10, no. 9, p. 1339, Aug. 2018.

[17] H. Zhang, K. Sun, and W. Li, "Object-oriented shadow detection and removal from urban high-resolution remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 11, pp. 6972–6982, Nov. 2014.

[18] L. Wang et al., "UNetFormer: A UNet-like transformer for efficient semantic segmentation of remote sensing urban scene imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 190, pp. 196–214, Aug. 2022.

[19] L. Ding, H. Tang, and L. Bruzzone, "LANet: Local attention embedding to improve the semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 426–435, Jan. 2021.

[20] C. Peng, K. Zhang, Y. Ma, and J. Ma, "Cross fusion net: A fast semantic segmentation network for small-scale semantic information capturing in aerial scenes," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5601313.

[21] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7132–7141.

[22] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 3–19.

[23] X. He, Y. Zhou, J. Zhao, D. Zhang, R. Yao, and Y. Xue, "Swin Transformer embedding UNet for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4408715.

[24] X. Wang, R. Girshick, A. Gupta, and K. He, "Non-local neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7794–7803.

[25] J. Fu et al., "Dual attention network for scene segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3146–3154.

[26] R. Li et al., "Multiattention network for semantic segmentation of fine-resolution remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5607713.

[27] M. Y. Yang, S. Kumaar, Y. Lyu, and F. Nex, "Real-time semantic segmentation with context aggregation network," *ISPRS J. Photogramm. Remote Sens.*, vol. 178, pp. 124–134, Aug. 2021.

[28] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. 18th Int. Conf. Med. Image Comput. Comput.-Assist. Intervent. (MICCAI)*, Munich, Germany. Cham, Switzerland: Springer, Oct. 2015, pp. 234–241.

[29] T. Y. Lin, P. Dollàr, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 2117–2125.

[30] T. Xiao, Y. Liu, B. Zhou, Y. Jiang, and J. Sun, "Unified perceptual parsing for scene understanding," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 418–434.

[31] L. Wang, R. Li, C. Duan, C. Zhang, X. Meng, and S. Fang, "A novel transformer based semantic segmentation scheme for fine-resolution remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[32] Q. Wang, B. Wu, P. Zhu, P. Li, W. Zuo, and Q. Hu, "ECA-Net: Efficient channel attention for deep convolutional neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 11534–11542.

[33] X. Ding, X. Zhang, J. Han, and G. Ding, "Scaling up your kernels to 31×31: Revisiting large kernel design in CNNs," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 11963–11975.

[34] L. C. Chen, G. Papandreou, and I. Kokkinos, "DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 834–848, Jun. 2017.

[35] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder–decoder with atrous separable convolution for semantic image segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 801–818.

[36] J. Wang et al., "Deep high-resolution representation learning for visual recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 10, pp. 3349–3364, Oct. 2020.

[37] C. Yu, J. Wang, C. Peng, C. Gao, G. Yu, and N. Sang, "BiSeNet: Bilateral segmentation network for real-time semantic segmentation," in *Proc. Eur. Conf. Comput. Vis.*, Sep. 2018, pp. 325–341.

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[38] B. Cheng, A. Schwing, and A. Kirillov, "Per-pixel classification is not all you need for semantic segmentation," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 17864–17875.
[39] L. Ding, J. Zhang, and L. Bruzzone, "Semantic segmentation of large-size VHR remote sensing images using a two-stage multiscale training architecture," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 8, pp. 5367–5376, Aug. 2020.
[40] S. Pan, Y. Tao, C. Nie, and Y. Chong, "PEGNet: Progressive edge guidance network for semantic segmentation of remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 4, pp. 637–641, Apr. 2021.
[41] Z. Cao et al., "End-to-end DSM fusion networks for semantic segmentation in high-resolution aerial images," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 11, pp. 1766–1770, Nov. 2019.
[42] Q. Zhao, J. Liu, Y. Li, and H. Zhang, "Semantic segmentation with attention mechanism for remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5403913.
[43] Z. Huang, X. Wang, L. Huang, C. Huang, Y. Wei, and W. Liu, "CCNet: Criss-cross attention for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 603–612.
[44] R. Niu, X. Sun, Y. Tian, W. Diao, K. Chen, and K. Fu, "Hybrid multiple attention network for semantic segmentation in aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5603018.
[45] X. Yang et al., "An attention-fused network for semantic segmentation of very-high-resolution remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 177, pp. 238–262, Jul. 2021.
[46] X. Li et al., "A synergistical attention model for semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5400916.
[47] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
[48] J. Dai et al., "Deformable convolutional networks," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 764–773.
[49] X. Zhu, H. Hu, S. Lin, and J. Dai, "Deformable ConvNets v2: More deformable, better results," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 9308–9316.
[50] J. Wang, Z. Zheng, A. Ma, X. Lu, and Y. Zhong, "LoveDA: A remote sensing land-cover dataset for domain adaptive semantic segmentation," 2021, *arXiv:2110.08733*.
[51] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," 2017, *arXiv:1711.05101*.
[52] Q. Liu, M. Kampffmeyer, R. Jenssen, and A.-B. Salberg, "Dense dilated convolutions' merging network for land cover classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 9, pp. 6309–6320, Sep. 2020.
[53] X. Zheng, X. Wu, L. Huan, W. He, and H. Zhang, "A gather-to-guide network for remote sensing semantic segmentation of RGB and auxiliary image," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5404915.
[54] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*.
[55] Z. Liu et al., "Swin Transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10012–10022.
[56] Q.-L. Zhang and Y.-B. Yang, "SA-Net: Shuffle attention for deep convolutional neural networks," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Jun. 2021, pp. 2235–2239.
[57] X. Li, W. Wang, X. Hu, and J. Yang, "Selective kernel networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 510–519.
[58] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 6000–6010.



Jia Liu (Member, IEEE) received the B.S. and Ph.D. degrees in electronic engineering from Xidian University, Xi'an, China, in 2013 and 2018, respectively.

He is currently an Associate Professor with the School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing, China. His research interests include computational intelligence and image understanding.



Wenyi Hua received the bachelor's degree in computer science and technology from Ningbo University, Ningbo, China, in 2022. He is currently pursuing the master's degree in computer technology with the Nanjing University of Science and Technology (NJUST), Nanjing, China.

His research interests include deep learning and remote sensing image segmentation.



Wenhua Zhang (Member, IEEE) received the B.S. degree from the North University of China, Taiyuan, China, in 2015, and the Ph.D. degree from Xidian University, Xi'an, China, in 2021.

She is currently a Post-Doctoral Researcher with the School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing, China. Her research interests include object tracking and image processing.



Fang Liu (Member, IEEE) received the B.S. degree in information and computing science from Henan University, Kaifeng, China, in 2012, and the Ph.D. degree in intelligent information processing from Xidian University, Xi'an, China, in 2018.

She is currently an Associate Professor with the Nanjing University of Science and Technology, Nanjing, China. Her research interests include deep learning, object detection, polarimetric synthetic aperture radar (SAR) image classification, and change detection.



Liang Xiao (Senior Member, IEEE) received the B.S. degree in applied mathematics and the Ph.D. degree in computer science from the Nanjing University of Science and Technology (NJUST), Nanjing, China, in 1999 and 2004, respectively.

From 2006 to 2008, he was a Post-Doctoral Research Fellow with the Pattern Recognition Laboratory, NJUST. From 2009 to 2010, he was a Post-Doctoral Fellow with the Rensselaer Polytechnic Institute, Troy, NY, USA. Since 2013, he has been the Deputy Director of the Jiangsu Key Laboratory of Spectral Imaging Intelligent Perception, Nanjing. Since 2014, he has been the Vice Director of the Key Laboratory of Intelligent Perception and Systems for High-Dimensional Information, Ministry of Education, NJUST, where he is currently a Professor with the School of Computer Science and Engineering. His research interests include remote sensing image processing, image modeling, computer vision, machine learning, and pattern recognition.

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